

OFFICIAL TECHNICAL SPECIFICATION

Software Requirements Specification (SRS)

SmartFarm Management Application

Version: 2.0 (Approved Baseline) **SDLC Stage:** Phase 1 — Requirements Specification & Compliance Expansion **Status:** Approved Baseline

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document details the complete functional, non-functional, security, and architectural requirements for the **SmartFarm Management Application**. It serves as the authoritative single source of truth for engineering, compliance testing, and continuous maintenance.

1.2 Scope of the System

SmartFarm is an enterprise-grade cloud farm management platform designed for modern agricultural enterprises. The application enables:

- Executive governance, labor auditing, and financial/debt oversight by **Farm Administrators**.
- Sector-level resource management, project planning, supplier procurement, and supervisor delegation by **Farm Managers**.
- Field-level task coordination, harvest logging, input consumption, and verified laborer work assignment by **Field Supervisors** under strict financial privacy isolation.
- Comprehensive tracking of external trade relationships: **Suppliers (Accounts Payable / Farm Debt)** and **Customers (Accounts Receivable / Customer Credit)**.
- Strict national labor compliance ensuring legal adult workforce verification (\$Age \ge 18\$) and zero tolerance for child labor.

1.3 Definitions, Acronyms, and Abbreviations

RBAC: Role-Based Access Control.

PBAC: Privilege/Policy-Based Access Control (fine-grained feature toggle permissions).

Labor Compliance: Strict validation of government National ID cards to guarantee legal adult employment eligibility (\$Age \ge 18\$) and eliminate child labor risks.

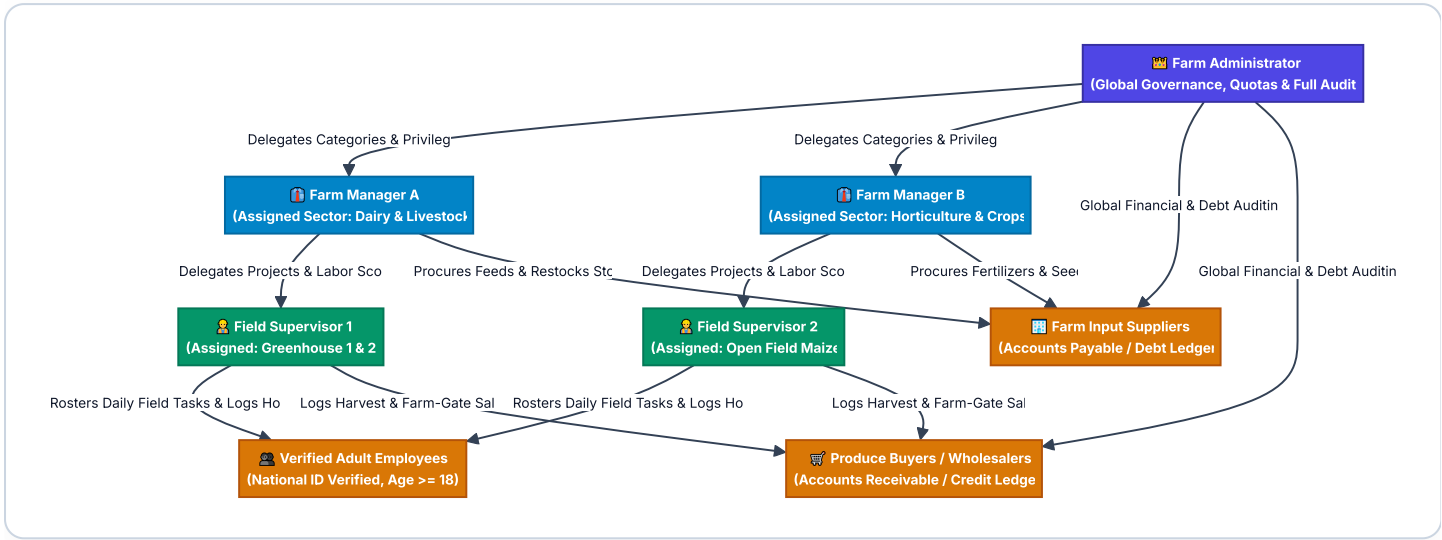
Accounts Payable (AP): Cumulative financial liabilities owed by the farm to external input/feed/equipment suppliers.

Accounts Receivable (AR): Cumulative credit balances owed to the farm by harvest and produce buyers.

Zero-State Dashboard: A tailored, clean UI rendered when an authenticated user has zero assigned projects/categories, suppressing empty graphs and displaying actionable next steps.

2. Overall Description & System Context

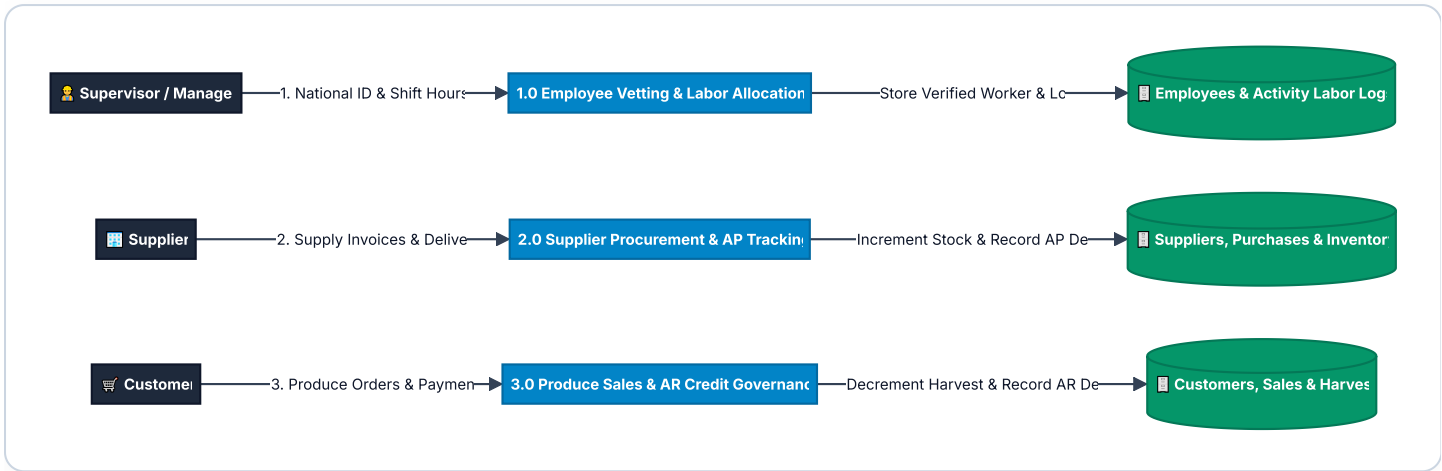
2.1 System Architecture & Actor Hierarchy



2.2 System Actors & Responsibility Matrix

Persona / Entity	System Scope	Max Quota	Core Responsibilities
Farm Administrator	Entire Farm (Global)	1 Account	Global user provisioning, AP/AR debt auditing, category allocation, system governance, and database oversight.
Farm Manager	Assigned Categories	Max 2 Accounts	Project budgeting, supervisor provisioning, supplier procurement, labor auditing, and sector reporting.
Field Supervisor	Assigned Projects	Max 10 Accounts	Daily field activity execution, harvest logging, worker task allocation, stock usage, and farm-gate sales under financial privacy isolation.
Employees / Laborers	Assigned Tasks	Scalable Registry	Verified adult workforce (\$Age \ge 18\$, National ID required) executing field operations with automated wage calculations.
Suppliers	Farm Inputs Catalog	Scalable Directory	Vendors providing seed, fertilizer, animal feed, tools, and machinery on cash or invoice credit terms.
Customers	Produce Distribution	Scalable Directory	Wholesale/retail buyers purchasing farm yields on cash, M-Pesa, or credit ledger terms.

3. Data Flow Diagram (DFD Level 1)



4. Detailed Functional Requirements (FR)

FR-1: Authentication, Account Provisioning & Password Recovery

FR-1.1: System shall support authentication using either `Username` or `Email` combined with a password.

FR-1.2: Unauthenticated users can request a password reset by providing their registered email address (`POST /users/forgot-password`).

FR-1.3: If an email is not registered in the system, the API shall return HTTP 400 with: `"No account found with this email address. If you did not register an email, please contact your Farm Administrator to reset your password."`

FR-1.4: If the email exists, the system generates a secure cryptographic token (`PasswordResetToken`) valid for 1 hour and sends an HTML reset email via SMTP.

FR-1.5: Administrators can execute direct password resets for staff accounts via the User Management portal (`PATCH /users/{id}/admin-reset-password`).

FR-2: Staff Quotas & Hierarchical Access Delegation (PBAC)

FR-2.1: The system shall strictly enforce staff limits: Max 1 Admin, Max 2 Managers, Max 10 Supervisors.

FR-2.2: Admin to Manager Delegation:

- Admin assigns one or more Farm Categories to a Manager.
- Admin configures fine-grained privilege toggles for the Manager (`CAN_CREATE_CATEGORIES` , `CAN_CREATE_SUPERVISORS` , `CAN_VIEW_FINANCIALS` , `CAN_MANAGE_BUDGETS` , `CAN_DELETE_INVENTORY`).

FR-2.3: Manager to Supervisor Delegation:

- Manager provisions supervisors linked to their manager account ID (`manager_id`).
- Manager assigns specific projects (within the Manager's categories) to the Supervisor.
- Manager configures fine-grained privilege toggles for the Supervisor (`CAN_RECORD_HARVEST` , `CAN_LOG_ACTIVITIES` , `CAN_USE_INVENTORY` , `CAN_RECORD_EXPENSES` , `CAN_RECORD_SALES`).

FR-3: Workload & Capacity Validation Rules

FR-3.1 (Manager Workload): When assigning a category to a Manager, if assignments exceed 3 active categories, an advisory workload confirmation is triggered.

FR-3.2 (Supervisor Workload): Each supervisor has a `max_project_capacity` (default 4 projects). The system blocks assigning additional active projects unless capacity is adjusted or existing projects complete.

FR-4: Category & Project Lifecycle Management

FR-4.1: Projects must belong to an existing Category and possess a globally unique project name.

FR-4.2 (Date Validation Rule): The system shall strictly validate that `startDate ≤ endDate` . If `startDate > endDate` , both backend and frontend reject the request with: `"Start date cannot be greater than end date!"`

FR-4.3 (Budget Modification Rule): Modifying project details, season, dates, or budgets strictly requires `CAN_MANAGE_BUDGETS` privilege.

FR-5: Operational Field Tracking (Harvest, Activity, Inventory)

FR-5.1: Harvest records capture item name, numerical yield, unit of measurement (KG, Litres, Bags, Crates), date, and notes.

FR-5.2: Activities capture activity title, type (Weeding, Irrigation, Spraying, Feeding, Pruning), date, notes, and attached laborers.

FR-5.3: Inventory tracks quantity in stock, unit prices, reorder levels, and records deduction history when consumed by projects.

FR-6: Financial Governance & Sales Enforcement

FR-6.1 (Stock-Backed Sales Rule): A project cannot record a sale exceeding its total recorded harvest yields minus previous sales.

FR-6.2 (Supervisor Financial Shield): Supervisor API endpoints and dashboard views strip budget allocations, sales revenue, and profit calculations.

FR-7: Dynamic Scoped Dashboards & Zero-State Engine

FR-7.1 (Admin Dashboard): Renders enterprise-wide aggregates, category financial distribution, user quotas, and global cashflow.

FR-7.2 (Manager Dashboard): Aggregates budget, project counts, and supervisor workloads strictly for the manager's assigned categories.

FR-7.3 (Supervisor Dashboard): Aggregates harvest volumes, pending tasks, and recent logs strictly for assigned projects.

FR-7.4 (Zero-State Suppression): If a user has 0 assigned categories or 0 assigned projects:

- All metrics cards, financial figures, and graphs are **suppressed**.
- Renders a dedicated **Action Hero Card** providing immediate contextual instructions (e.g., `"No categories assigned yet. Contact your Farm Administrator to get started."`).

FR-8: User Deletion & Referential Integrity Safeguards

FR-8.1: Admin accounts cannot be deleted (`isUserAdmin` protection).

FR-8.2: Deleting a user safely cascades/unlinks child records: deletes active `PasswordResetToken` rows, clears category join tables, unlinks `supervisor_id` on projects, and clears `createdById` references.

FR-9: Labor Compliance, Legal Adult Verification & Work Assignment



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FR-9.1 (Adult Labor Verification): All farm hands and field workers must be registered with their official government **National ID number**, **Full Name**, and **Phone Number** (for identification and M-Pesa payroll).

FR-9.2 (Minor Employment Prohibition): System strictly blocks onboarding laborers without verified government identity credentials, preventing child labor compliance violations.

FR-9.3 (Employment Classification): Supports **CASUAL** (daily wage / task-based) and **PERMANENT** (salaried / monthly) employee profiles.

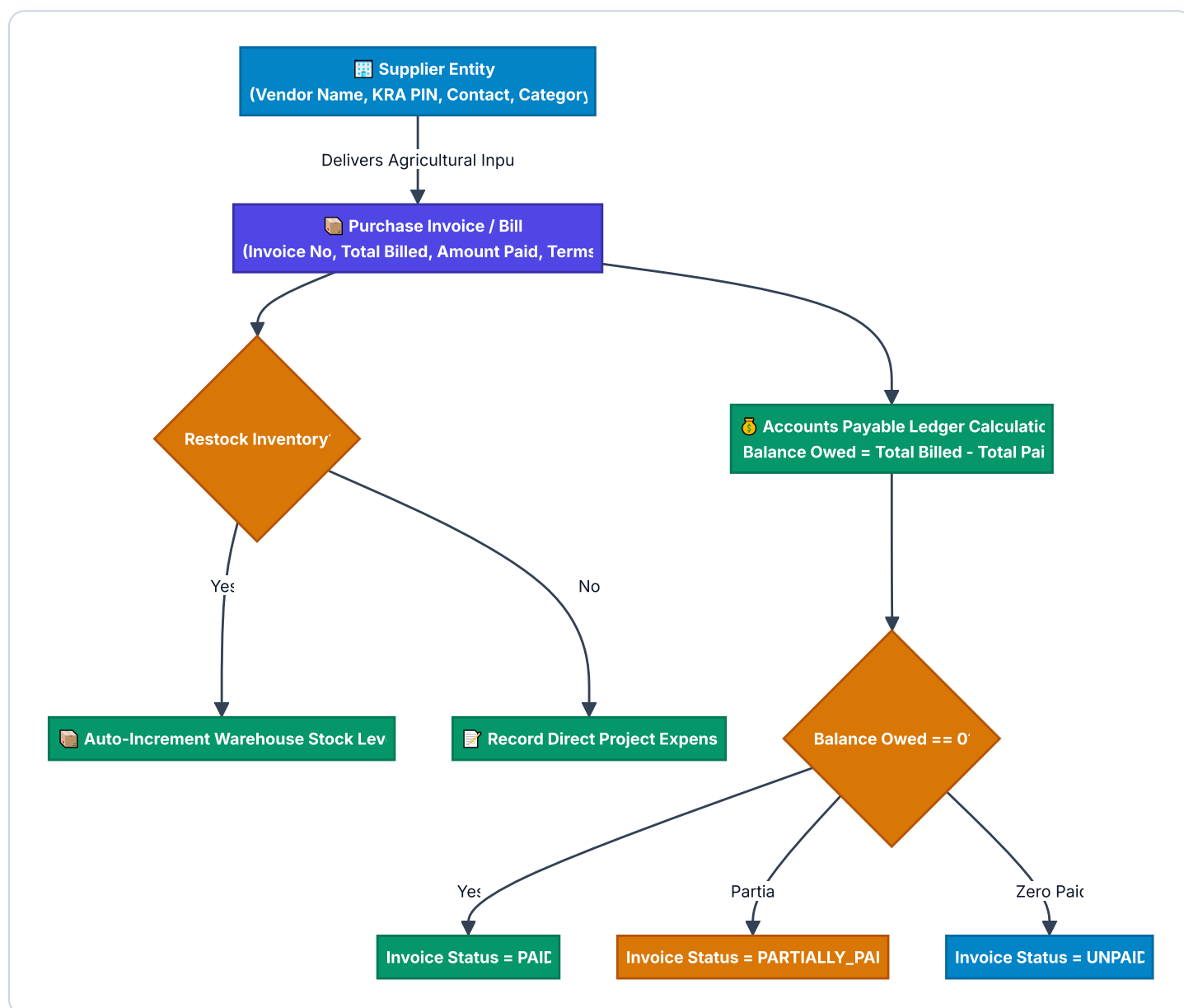
FR-9.4 (Task & Activity Allocation):

- Admins, Managers, and Field Supervisors can assign one or more verified employees to any project activity (e.g., *Weeding Sector 1*, *Spraying Block C*, *Irrigation Line Setup*).

- Captures: **activity_id**, **employee_id**, **date**, **hours_worked**, **daily_rate**, **wage_payable**, and supervisor remarks.

FR-9.5 (Labor Audit Trail & Payroll Summary): Provides a complete historical log of all tasks performed by an employee across farm projects for labor auditing and wage settlements.

FR-10: Supplier Management & Accounts Payable (Debt) Ledger



FR-10.1 (Supplier Registry): Centralized directory of external farm suppliers capturing: **name**, **contact_person**, **phone_number**, **email**, **id_or_tax_number** (KRA PIN / National ID), **category_supplied** (*Fertilizer & Agro-chemicals*, *Animal Feeds*, *Seeds & Seedlings*, *Farm Machinery & Hardware*, *Veterinary Supplies*), and physical address.

FR-10.2 (Purchase & Expense Linkage): When stock is added to inventory or project cash expenses are recorded, transactions can be linked directly to a registered `supplier_id`.

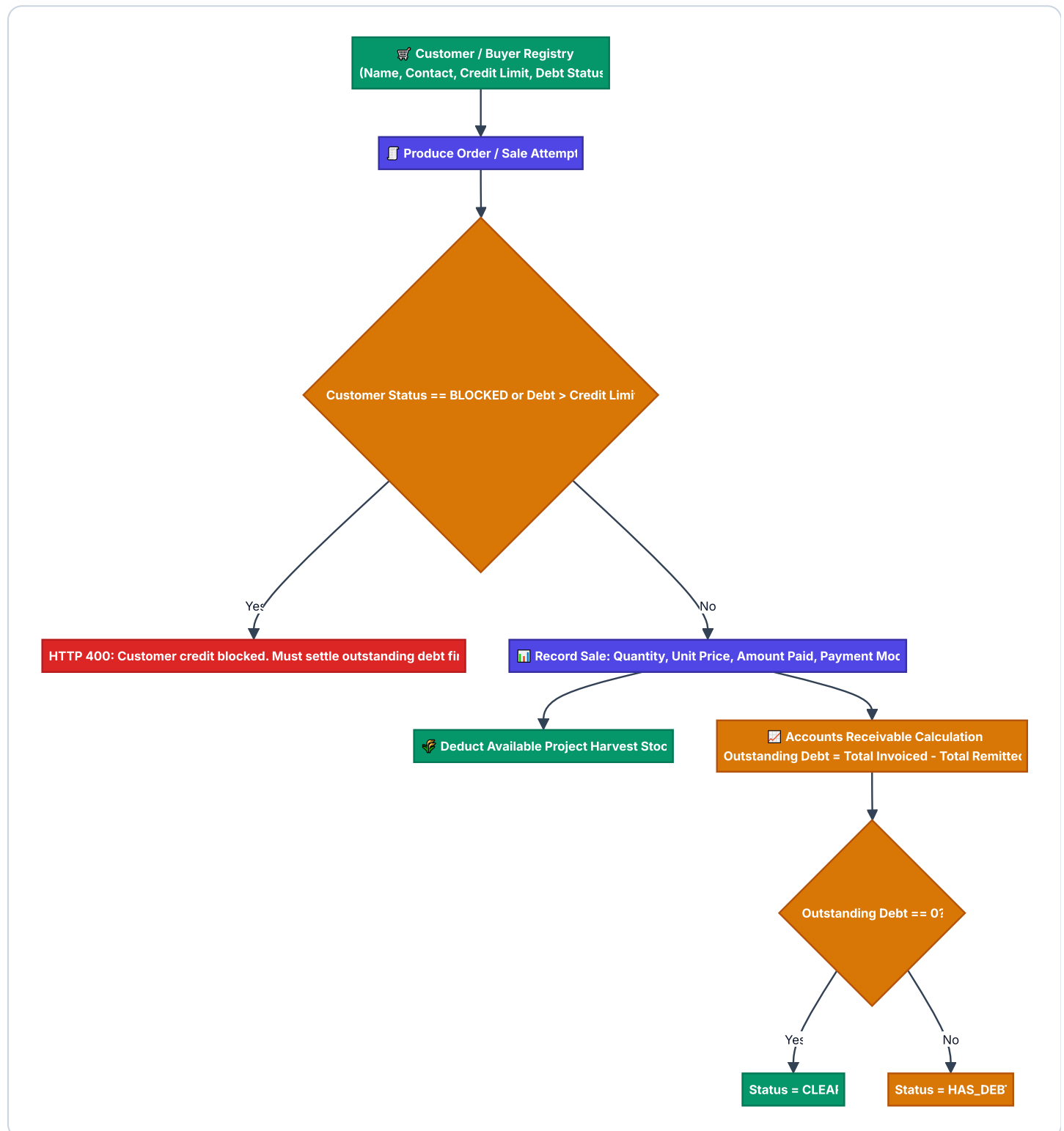
FR-10.3 (Accounts Payable / Debt Tracking):

- For each supplier transaction, captures: `invoice_number`, `invoice_amount`, `amount_paid`, `balance_due`, `payment_status` (`PAID`, `PARTIALLY_PAID`, `CREDIT_UNPAID`), and `payment_due_date`.

- Calculates live supplier balances: $\text{Balance Owed} = \sum \text{Invoice Amount} - \sum \text{Amount Paid}$.

FR-10.4 (Executive Debt Governance): Admin and authorized Managers have dedicated dashboard widgets tracking Total Farm Accounts Payable (Outstanding Debt Owed to Suppliers) with overdue liability alerts.

FR-11: Customer Credit Governance & Accounts Receivable Ledger



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- FR-11.1 (Customer Registry Expansion):

Stores buyer identity including `name` , `contact` , `id_number` , `address` , `credit_limit` , and `credit_status` (`CLEAR` , `HAS_DEBT` , `CREDIT_BLOCKED`).
- FR-11.2 (Credit Sales & Partial Payments):

- When recording sales, supports immediate full cash/M-Pesa payment, partial payment, or full credit terms.

- Captures: `total_amount` , `amount_paid` , `balance_due` ($\text{total_amount} - \text{amount_paid}$), `payment_mode` (`CASH` , `MPESA` , `BANK_TRANSFER` , `CREDIT_LEDGER`), and `payment_status` (`PAID_IN_FULL` , `PARTIAL_PAYMENT` , `CREDIT_UNPAID`).
- FR-11.3 (Accounts Receivable / Debtor Tracking):

- Automatically calculates customer debt balances: $\text{Outstanding Debt} = \sum \text{Total Invoiced} - \sum \text{Total Remitted}$.

- Flags customers exceeding credit terms or with overdue balances during new sale attempts.
- FR-11.4 (Supervisor Privacy Isolation):

Field Supervisors can log transaction-level cash/credit splits for direct farm-gate sales without gaining access to cumulative farm-wide debtor reports.

5. Non-Functional Requirements (NFR)

- NFR-1 (Security):

BCrypt password hashing ($\text{cost} = 10$), role/privilege-based endpoint security, parameter sanitization, and SQL injection prevention via Spring Data JPA.
- NFR-2 (Performance):

Page load times $< 1.5\text{s}$ on broadband; database indexed on `username` , `email` , `category_id` , `supervisor_id` , `manager_id` , `employee_id` , `supplier_id` , `customer_id` .
- NFR-3 (Reliability):

99.9% uptime target backed by automated heartbeat monitoring.
- NFR-4 (Cost):

100% compliant with zero-cost cloud architecture limits (Render Free Web Service, Vercel Hobby, TiDB Cloud / Aiven Free Tier).
- NFR-5 (Usability):

Modern design system with responsive layouts, accessible color contrast, and dedicated mobile field data entry views.